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Paper Code : CS-601

DATABASE MANAGEMENT SYSTEM

Time Allotted: 3 Hours | *Full Marks:* 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A

(Multiple Choice Type Questions)

1. Choose the correct alternative for the following: 1×10=10

(i) Which of the following is not a property of transaction?
(a) Atomicity (b) Concurrency
(c) Isolation (d) Durability

(ii) Check points are part of
(a) Recovery measure (b) Security measure
(c) Concurrency measure (d) Authorization measure

(iii) Tree structures are used to store data in
(a) Network model (b) Relational model
(c) Hierarchical model (d) File based system

(iv) The language that requires a user to specify the data to be retrieved without specifying how to get it is
(a) Procedural DML (b) Non procedural DML
(c) Procedural DDL (d) Non-procedural DDL

(v) The rule that a value of a foreign key must appear as a value of some specific table is called a
(a) Referential constraint (b) Index
(c) Integrity constraint (d) Functional Dependency

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(vi) The clause in SQL that specifics that the query result should be sorted in ascending or descending order based on the values of one or more column is

(a) view (b) order by

(c) group by (d) having

(vii) Conflict serializability can be detected by

(a) WFG (b) precedence graph

(c) spanning tree (d) None of these

(viii) According to the levels of abstractions the schema at the intermediate level is called

(a) Logical schema (b) Physical schema

(c) Subschema (d) Super schema

(ix) The operation which is not considered as basic operation in relational algebra

(a) Join (b) Selections

(c) Union (d) Cross product

(x) It is an abstraction through which relationships are treated as higher level entities?

(a) Generalisation (b) Specialisation

(c) Aggregation (d) Inheritence

Group - B

(Short Answer Type Questions)

Answer *any three* of the following: 5×3=15

2. Describe different 2PL protocol in brief.

3. Consider the following database with primary keys underlined

Project (P - No, P - Name, P - incharge)

Employee (E - no, E - name)

Assigned –To (E – no, P - no)

Write relational algebra expression for the following:

(a) List detail of employee working on all projects.

(b) List E - no, E - name of employee who do not work on project with P - no = DB2003. 2½+2½=5

4. What is partial functional dependency? Explain BCNF with a suitable example. 2+3=5

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5. Consider the following two transactions:

T1 : read (A); read (B); if A = 0, then B = B + 1; write (B);

T2 : read (B); read (A); if B = 0, then A = A + 1; write (A);

Add lock and unlock instructions to transaction T1 and T2. Is there any dead lock observed?

6. Explain Armstrong axiom in brief.

Group – C

(Long Answer Type Questions)

Answer *any three* of the following: 15×3=45

7. (a) Why do we need normalisation?

(b) What are the advantages of database over traditional file system?

(c) Explain the role of database administrator.

(d) Relation assignment (worker-id, building -id, start-date, name, skill).

FD's are – {worker-id → name, {worker-id, building-id} → start date, name → skill}. Is the relation in 3 NF, if not then make it in 3NF. http://www.makaut.com 3+4+3+5=15

8. (a) What is the utility of Query Tree?

(b) Describe different phase of a Query Tree.

(c) Consider the employee query "Select work-on, e-name, from employee, proj-assign project where p-name = "ASSEMBLY" and p-no = "p1" and Join-Date = "21-12-17" and employee. eno = proj-assign. eno and proj-assign, pno = project. p-no.

(d) "Primary keys are candidate keys but reverse is not true."—explain. 3+4+6+2=15

9. (a) What is deadlock? What are necessary and sufficient condition for deadlock?

(b) What are the necessary measurement for deadlock avoidance?

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(c) Let T_1, T_2 and T_3 be transactions that operate on the same data items A, B, and C. Let $r_1(A)$ mean that T_1 read A, W_1 (A) means that T_1 writes A and so on for T_2 and T_3. Consider the following schedule:

$S_1 : r_2(C), r_2(B), W_2(B), r_3(B), r_3(C), r_1(A)$ $W_1 : W_3(B), W_3(C), r_2(A), r_1(B), W_1(B), W_2(A)$

Is the schedule serializable? (1+4)+3+7=15

10. Answer the following queries in SQL using given database schema:

EMP (E no, E name, E add, B date, super-no)

DEPT (D no, D name, mgrno)

PROJECT (P no, P name, D no, Plocation)

Work-on (E no, D no, Hours)

(a) List E no, E name, E add, mgrno of all employee who work in "Research" department.

(b) For all project in "Kolkata" list p-no, controlling department name, manager name, address and birth date.

(c) List all project number, project name, manager name which belongs to "Product" department.

(d) List all employee name who works more than 40 hours in "Research" department. 4+4+4+3=15

11. Answer *any three* questions: 5×3=15

(a) Time stamp based protocol for concurrency control

(b) B+ tree file organisation

(c) Primary Index and Secondary Index

(d) Anomalies in concurrent execution of transaction.

(e) BCNF

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